

MUHAMMAD QAIM ALI

AI & Machine Learning Engineer

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PROFESSIONAL SUMMARY

AI and Machine Learning Engineer with a proven track record of delivering production-grade systems across healthcare, manufacturing, and automation. Architected end-to-end ML pipelines from data ingestion to cloud deployment, leveraging deep expertise in LLMs, computer vision, and MLOps. Consistently recognized for academic excellence and client impact, combining research rigor with engineering precision to build scalable, real-world AI solutions.

TECHNICAL SKILLS

Programming: Python, C++, SQL **ML/DL:** PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, LoRA/PEFT, Model Optimization
NLP & LLMs: LangChain, spaCy, NLTK, RAG, Vector Databases (Qdrant, pgvector), Prompt Engineering
Computer Vision: YOLOv8, ResNet, DICOM Processing, 3D Vision-Language Models, OpenCV
MLOps & Data Engineering: MLflow, DVC, Apache Airflow, Docker, Docker Compose, CI/CD (GitHub Actions)
Cloud (AWS): EC2, RDS, S3, IAM, VPC, ASG, ELB, CloudWatch **Databases:** PostgreSQL (pgvector), Neo4j, Qdrant, MongoDB
APIs & Backend: FastAPI, RESTful APIs, JWT, RBAC, Async SQLAlchemy **Tools:** VS Code, Git, Docker, Jupyter

EDUCATION

FAST National University of Computer & Emerging Sciences (NUCES) Faisalabad, Pakistan
B.S. Artificial Intelligence — CGPA: 3.14 2022 - 2026

- Relevant Coursework: Artificial Intelligence, Machine Learning, Deep Learning, NLP, Computer Vision, MLOps, Cloud Computing, Data Structures & Algorithms, Database Systems
- Silver Medalist — 1st Semester (Fall 2022); DEAN's List — 7th Semester (Fall 2025); DEAN's List — 8th Semester (Spring 2026)

EXPERIENCE

Arbisoft Islamabad, Pakistan
Machine Learning Intern Jul 2025 – Sep 2025

- Contributed to *Alithia Knowledge*, an LLM-based intelligent knowledge retrieval system serving enterprise clients.
- Reduced query latency by migrating the knowledge base from Neo4j to Qdrant Vector DB, significantly improving retrieval throughput; validated embedding pipelines for semantic consistency with zero data loss.

Fiverr Remote
Python, Selenium & ML Freelancer Mar 2024 – Present

- Delivered 100+ Python-based projects with 5-star satisfaction, spanning browser automation (Selenium), custom extensions, and end-to-end ML/DL pipelines for clients across e-commerce, healthcare, and finance.

FAST-NUCES Faisalabad, Pakistan
Teaching Assistant Aug 2023 – Jun 2026

- Courses: Operating Systems, Expository Writing, English Composition & Comprehension

PROJECTS

PAB MRI: Multimodal AI for Structured Clinical Report Generation | *PyTorch, LoRA, DICOM, MLflow* 2025–2026

- Engineering a 3D Vision-Language Model pipeline converting raw MRI scans into structured clinical narratives, targeting radiologist-level accuracy on a proprietary dataset of 4,000+ patient scans with full IRB-compliant DICOM preprocessing.
- Applying LoRA fine-tuning within a full MLOps workflow (DVC, MLflow) for real-time diagnostic inference.

Intelligent Clinic System | *FastAPI, PostgreSQL, Scikit-learn, RAG, Docker, AWS* 2026

- Architected a production-grade multi-tenant healthcare API with JWT, RBAC, and Row-Level Security; trained an Isolation Forest model on 5,000 vital records achieving 100% precision/recall.
- Integrated a RAG pipeline (LLaMA 3, nomic-embed-text) for medical Q&A; containerized with Docker Compose and shipped with 49 automated tests via GitHub Actions CI/CD.

Manufacturing Defect Detection | *PyTorch, ResNet-50, Computer Vision* 2025

- Designed a ResNet-50 defect detection system with weighted loss strategies to handle class imbalance, improving F1-score by 18% over baseline across a full augmentation-to-deployment pipeline.